

Q5(a)(i)

Consider a line between P and R. When the disc spins, the line sweeps through the region of the magnetic field, cutting the magnetic flux lines.

By Faraday's Law, the change of magnetic flux with time results in an emf appearing between points P and R.

Q5(a)(ii)

Assume that the magnetic field is uniform across the region of PR.

Using emf $\varepsilon = B\pi r^2 f$

$$\frac{\varepsilon_1}{\varepsilon_2} = \left(\frac{r_1}{r_2}\right)^2$$
$$\varepsilon_2 = \left(\frac{0.3r}{r}\right)^2 \varepsilon_1 = \mathbf{0.90 \text{ mV}}$$

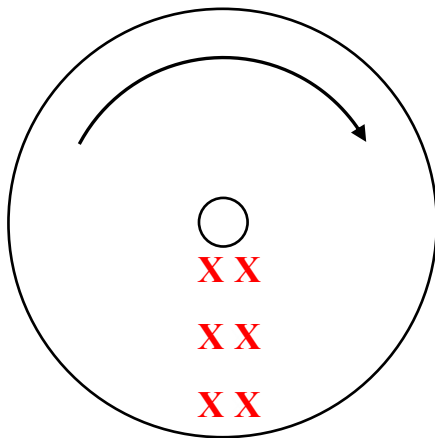
Q5(a)(iii)

Because $\varepsilon_{QR} = 0.90 \text{ mV}$ and since
 $\varepsilon_{PQ} = 10 - 0.90 = \mathbf{9.1 \text{ mV}}$

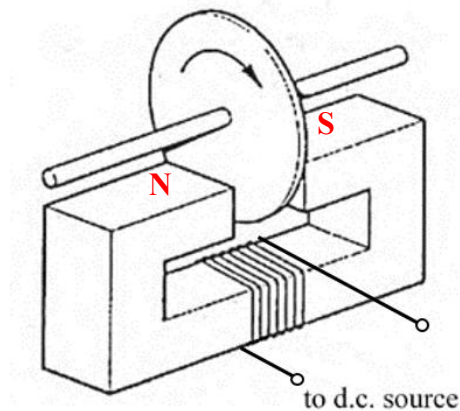
$$\varepsilon_{PR} = 10.0 \text{ mV},$$

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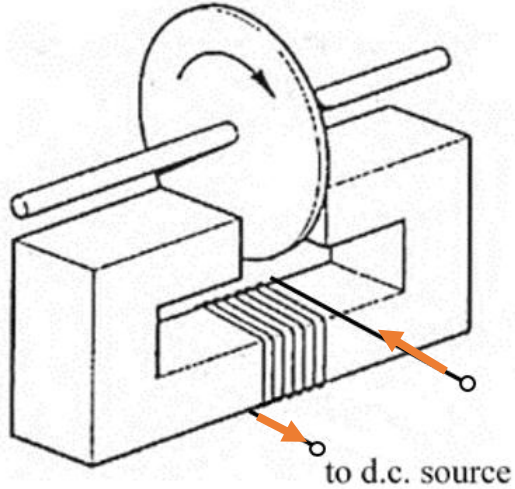
Q5(b)(i)



5(b)(ii)



5(b)(iii)



Q5(c)

When the flux density is over the whole disc, the emf generated between the points PR will again be 10.0 mV.

Q6(a)